Statistics 2103 Exam 1
Fall 2026

Name _____

Uni _____

The time limit for this exam is 75 minutes. Read each part of each question carefully. For open response questions, students must include all relevant work to receive full credit. This exam has 7 pages and is worth a total of 50 points.

One double-sided 8.5x11 inch sheet of notes is permitted. Non-graphing calculators are permitted. No other outside material or assistance is permitted.

Sign your name below to acknowledge your understanding of these rules.

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Section 1: True or false? Circle the appropriate choice (1 point each).

1. (TRUE / FALSE) A correlation of zero between two variables, X and Y, suggests that there is no relationship between X and Y.

2. (TRUE / FALSE) If two random variables X and Y are positively correlated, then $\text{var}(X+Y) > \text{var}(X) + \text{var}(Y)$.

3. (TRUE / FALSE) If 90% of all New Yorkers wear seat belts while driving, then the probability that 8 out of 10 randomly sampled New York drivers wear seat belts is equal to $0.9^8 * 0.1^2$.

4. (TRUE / FALSE) When we test a hypothesis about the value of a population mean μ, if the null hypothesis is true, the p-value for the test may take any value from 0 to 1.

5. (TRUE / FALSE) When constructing a confidence interval for a population mean μ, a lower confidence level results in a narrower interval, all else being equal.

6. (TRUE / FALSE) In a simple linear regression model, the residual for observation i is $y_i - (\beta_0 + \beta_1 x_i)$.

7. (TRUE / FALSE) The presence of a positive linear association between variables X and Y in observational data does not necessarily mean that increasing X will result in an increase in Y.

8. (TRUE / FALSE) For a simple linear regression model, it is always the case that $\sum_{i=1}^{n}(y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i))^2 \geq \sum_{i=1}^{n}(y_i - (\beta_0 + \beta_1 x_i))^2$.

9. (TRUE / FALSE) For the simple linear regression model, a useful tool for assessing the normality assumption is a histogram of the response variable Y. If the assumption is satisfied, the histogram should closely resemble the curve for a normal distribution.

10. (TRUE / FALSE) In a simple linear regression model, the Y value predicted by the least squares regression line for an observation with $X = \bar{x}$ is equal to the sample mean of the observed Y values.

Section 2: Circle the appropriate choice (2 points each).

11. If X is a continuous random variable, then the probability $\mathbb{P}(X = 4)$ is:

    (a) the height of the probability density function curve at $x = 4$

    (b) the area under the probability density function curve where $x < 4$

    (c) the area under the probability density function curve where $x > 4$

    (d) equal to zero

12. Suppose we want to estimate μ, the true average monthly rent of one-bedroom apartments in Chicago. Based on a random sample of apartments, we obtain a 95% confidence interval of \$1850 to \$2300. Which of the following is the best interpretation of that interval?

    (a) 95% of all one-bedroom Chicago apartments cost between \$1850 and \$2300 per month

    (b) the true average monthly rent is between \$1850 and \$2300 with 95% probability

    (c) we are 95% confident the true average monthly rent is between \$1850 and \$2300

    (d) none of the above

13. All else equal, which of the following does NOT tend to make the least squares estimator of the slope in a linear regression model have lower variance?

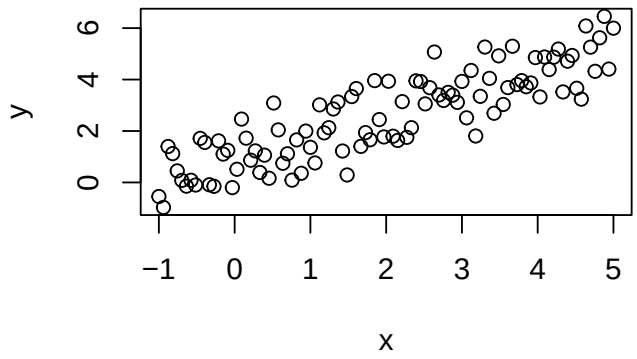
    (a) larger sample size

    (b) larger vertical spread of Y values around the line

    (c) larger variation in observed X values

    (d) none of the above

14. Suppose we fit a linear regression model to the data depicted in the scatter plot below, using X as the predictor and Y as the response.



For X values that differ by 1 standard deviation, we would predict Y values to differ by:

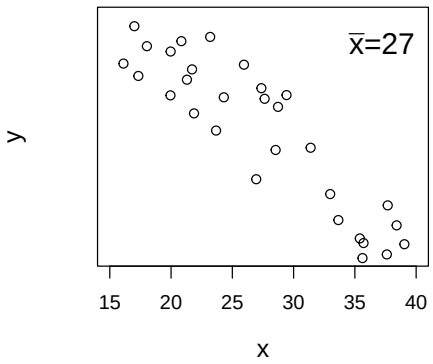
(a) less than one standard deviation

(b) exactly one standard deviation

(c) more than one standard deviation

(d) cannot be determined with the information given

Questions 15-17 concern a least squares regression line fit to data depicted in the scatter plot below.



15. Which of the following will be wider:

    (a) 95% confidence interval for the average Y value when X=40

    (b) 95% confidence interval for the average Y value when X=25

    (c) cannot be determined with the information given

16. Which of the following will be wider:

    (a) 95% confidence interval for the average Y value when X=30

    (b) 99% prediction interval for an individual Y value when X=30

    (c) cannot be determined with the information given

17. A simple linear regression model fit to the data depicted in the scatter plot above has $R^2 = 0.81$. Which of the following are possible values of the sample correlation between x and y?

    (a) 0.656 only

    (b) 0.90 and -0.90

    (c) -0.90 only

    (d) cannot be determined with the information given

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Section 3: Answer all questions in the space provided.

1. (4 points) A grocery store sells blocks of cheese with a mean weight of 1 lb and a standard deviation of 0.2 lb. You independently select two blocks. If the cheese costs \$12 per pound, what are the expected value and standard deviation of the total cost?

   $\mathbb{E}(12X_1 + 12X_2) = 12\mathbb{E}(X_1) + 12\mathbb{E}(X_2) = 12 * 1 + 12 * 1 = \$24$

$$\begin{aligned} \text{var}(12X_1 + 12X_2) &= \text{var}(12X_1) + \text{var}(12X_2) \\ &= 12^2\text{var}(X_1) + 12^2\text{var}(X_2) \\ &= 12^2 * 0.2^2 + 12^2 * 0.2^2 \end{aligned}$$

   $\text{var}(12X_1 + 12X_2) = \sqrt{12^2 * 0.2^2 + 12^2 * 0.2^2} \approx \$3.39$

2. (4 points) A chef conducts an experiment, baking identical cheesecakes at temperatures ranging from 300°F to 350°F and recording how long they take to finish baking. Most baking times were between 75 and 90 minutes; a linear model fitted to these data predicts that a cheesecake baked at 600°F would finish baking in less than one minute. Is that a reasonable prediction? Explain your answer from a statistical standpoint.

   No. A temperature of 600°F is very far outside the observed range of temperatures, so it isn't reasonable to assume the association between temperature and baking time would be the same.

3. (6 points) In 2010, 20% of 40-year-olds had never been married. In a sample of 62 40-year-olds surveyed in 2026, 18 had never been married. At an $\alpha = 0.05$ significance level, perform an appropriate hypothesis test to assess whether the current population proportion of 40-year-olds who have never been married exceeds 20%. Clearly state your conclusion in the context of this problem.

   This is asking for a test of $H_0 : p = 0.2$ vs. $H_a : p > 0.2$

$$z = \frac{(18/62) - 0.2}{\sqrt{0.2 * 0.8/62}} \approx 1.8$$

   This is greater than 1.64, so at the $\alpha = 0.05$ level, we reject H_0 and conclude the current proportion of never-married 40-year-olds IS greater than 20%.

4. (8 points) The output below shows the results of fitting a linear regression model to predict systolic blood pressure (in mmHg) from resting heart rate (in beats per minute).

```
lm(formula = bloodpressure ~ heartrate)
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)      95.2    4.93      19.31    <2e-16
heartrate        0.25    0.11       2.19     0.031
R-squared:  0.07
```

   (a) (3 points) What is the estimated average blood pressure among people with heart rates of 80 beats per minute?

   $\hat{y} = 95.2 + 0.25 * 80 = 95.2 + 20 = 115.2$mmHg

   (b) (3 points) State the interpretation of the value 0.25 in the output above.

   There are many correct ways to word the answer to this question. Responses should not imply that increasing the heart rate increases the blood pressure. Including units in responses is not required.

   Here are some good and not-so-good examples of how you could describe the interpretation.

   Good:

   - for people whose resting heart rates differ by 1 bpm, we expect their blood pressures to differ by 0.25mmHg on average
   - for people whose resting heart rates differ by 1 bpm, we predict the person with the faster heart rate to have higher blood pressure by 0.25 units
   - a 1 bpm difference in heart rate is associated with a 0.25 unit difference in average blood pressure
   - we estimate that the true average difference in blood pressure between people with heart rates that differ by 1 bpm is 0.25 mmHg

   Not-so-good

   - when heart rate increases by 1 bpm, blood pressure increases by 0.25mmHg
   - a 1 bpm increase in heart rate is associated with an 0.25 unit increase in blood pressure, on average

   (c) (2 points) Is there significant evidence of a linear association between heart rate and blood pressure? Justify your response based on the output above.

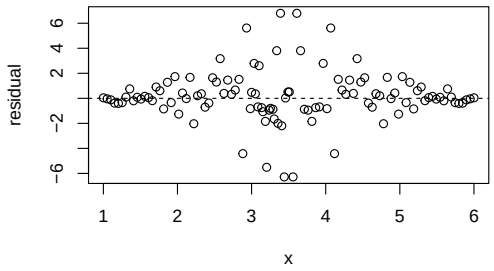
   Yes. From the summary table, you could point out either:

   - the p-value for a test of $H_0 : \beta_1 = 0$ vs. $H_a : \beta_1 \neq 0$ is approximately 0.031, so we would reject the null hypothesis that $\beta_1 = 0$ at an $\alpha = 0.05$ level

– the estimated slope was more than 2 standard deviations away from zero (technically since we don't know the sample size, this on its own would not tell you if the evidence is "significant" at the $\alpha = 0.05$ level, but I would accept this response anyway since the p-value is also provided and is small)

Note: the small R^2 value suggests that the association is weak/small in magnitude. However, we still have significant evidence that an association *exists* ("*is* there an association?" and "is the association *large*?" are two different questions).

5. (4 points) The plot below shows the residuals from a fitted simple linear regression model.



(a) (2 points) Based on this plot, which regression assumptions appear to be violated, if any?

error terms appear to have non-constant variance

(no other violations are apparent in this plot)

(b) (2 points) For the violated assumptions identified above (if any), what are the consequences for estimation of the linear model coefficients, and for inference (confidence intervals and hypothesis tests) based on the model?

Estimated coefficients are still valid/unbiased/correct (any of those terms is acceptable).

Inference based on the model will NOT be valid/correct.

Percentiles of Normal(0,1) distribution

$\mathbb{P}(Z \leq z)$	0.5	0.75	0.9	0.95	0.975	0.99	0.995
z	0	0.67	1.28	1.64	1.96	2.33	2.58

Percentiles of t-distribution

Degrees of freedom:	0.50	0.90	0.95	0.975	0.99	0.995
1	0.00	3.08	6.31	12.71	31.82	63.66
5	0.00	1.48	2.02	2.57	3.36	4.03
10	0.00	1.37	1.81	2.23	2.76	3.17
20	0.00	1.33	1.72	2.09	2.53	2.85
40	0.00	1.30	1.68	2.02	2.42	2.70
50	0.00	1.30	1.68	2.01	2.40	2.68
60	0.00	1.30	1.67	2.00	2.39	2.66